# FINAL DOCUMENTATION

## Task: Standards Study & Engineering Rule Extraction

# 1. Executive Summary

The objective of this task was to study relevant international industrial standards and extract structured engineering rules to support automated interpretation of Piping and Instrumentation Diagrams (P&ID) using Generative AI.

The standards reviewed include:

- ISA-5.1 – Instrumentation Symbols and Identification
- ISO 10628 – P&ID Diagrammatic Conventions
- IEC 61511 – Safety Instrumented Systems

The outcome of this task includes:

- A structured Standards Summary Document
- A machine-readable Rule Reference Sheet (Excel-based knowledge base)
- A deterministic engineering rule framework for AI integration

This work ensures that AI-based interpretation remains aligned with recognized industry engineering logic and safety principles.

# 2. Standards Studied and Key Extracted Concepts

## 2.1 ISA-5.1 – Instrumentation Symbols and Identification

ISA-5.1 defines standardized conventions for:

- Instrument tag structure
- Letter coding systems
- Functional identification
- Modifier conventions

### Extracted Core Rule

Instrument Tag Format:

[Process Variable][Function][Modifier]

Examples:

- PT → Pressure Transmitter
- FT → Flow Transmitter
- LSHH → Level Switch High-High
- PSLL → Pressure Switch Low-Low

### Engineering Derivation

From ISA-5.1, the following signal classification logic was derived:

- Transmitters → Analog Input (AI)
- Switches → Digital Input (DI)

- Control Valves → Analog Output (AO)
- Shutdown Valves → Digital Output (DO)

These deterministic mappings directly support automated I/O list generation and structured tag parsing.

# 2.2 ISO 10628 – P&ID Layout Conventions

ISO 10628 defines graphical and structural representation of:

- Process lines
- Signal lines
- Equipment blocks
- Flow direction
- Isolation boundaries

### Extracted Structural Rules

- Main process lines are represented thicker than signal lines
- Dashed lines indicate control or signal connections
- Process flow is generally oriented left-to-right
- Isolation breaks define logical and operational sections

### Engineering Implication

These layout principles enable:

- Structural graph extraction
- Functional area segmentation
- Context-aware instrument mapping

# 2.3 IEC 61511 – Safety Instrumented Systems

IEC 61511 governs:

- Safety Instrumented Functions (SIF)
- Trip logic
- Alarm classification
- Shutdown conditions

## Extracted Safety Principles

Alarm:

- Indicates abnormal condition
- Requires operator intervention

Trip:

- Initiates automatic protective action
- Triggers equipment shutdown or isolation

## Derived Trip Logic Examples

- HH + SDV → Trip
- LL + Pump → Trip
- H only → Alarm

These deterministic rules prevent unsafe AI inference and ensure compliance with safety engineering principles.

# 3. Rule Extraction Methodology

The rule extraction process followed a structured analytical approach rather than direct transcription of standards.

The methodology included:

1. Identification of recurring engineering patterns
2. Extraction of tag decoding logic
3. Mapping device functions to signal types
4. Defining deterministic alarm and trip classifications
5. Structuring rules into a machine-readable format

The objective was not regulatory documentation, but practical engineering logic extraction for AI-based automation.

# 4. Structured Rule Reference Sheets

## Sheet 1 – Instrument_Rules

- Decode tag prefixes
- Identify process variable
- Identify function
- Assign signal type

| Tag | Variable | Function | Signal |
|---|---|---|---|
| PT | Pressure | Transmitter | AI |
| PSHH | Pressure | Switch High-High | DI |
| FCV | Flow | Control Valve | AO |
| SDV | Safety | Shutdown Valve | DO |

Role in AI:

- Enables NLP-based tag parsing
- Supports automatic I/O classification
- Forms the foundation for structured interpretation

# Sheet 2 – Signal_Mapping

- Map device types to PLC/DCS signal classifications

| Device | Signal Type |
|---|---|
| Transmitter | AI |
| Switch | DI |
| Control Valve | AO |
| Shutdown Valve | DO |

Role in AI:

- Enables automatic generation of I/O lists
- Supports hardware-level mapping

# Sheet 3 – Alarm_Rules

- Classify alarm severity based on modifier codes

| Tag | Variable | Function | Signal |
|---|---|---|---|
| PT | Pressure | Transmitter | AI |
| PSHH | Pressure | Switch High-High | DI |
| FCV | Flow | Control Valve | AO |
| SDV | Safety | Shutdown Valve | DO |

Role in AI:

- Enables automated alarm list generation
- Differentiates warning from critical conditions

# Sheet 4 – Trip_Rules

- Define automatic shutdown logic

| Trigger | Element | Action |
|---|---|---|
| PSHH | SDV | Shutdown |
| LSLL | Pump | Stop Pump |
| Emergency Stop | Entire Unit | Full Shutdown |

Role in AI:

- Enables automatic Trip List generation
- Validates GenAI reasoning against deterministic safety rules

# 5. Integration with AI Architecture

The extracted rule sheets support multiple layers of the AI system:

## 5.1 Tag Parsing Layer

ISA-based decoding ensures structured and consistent interpretation of instrument tags.

## 5.2 I/O List Generation

Signal mapping rules convert instruments into AI, AO, DI, and DO classifications.

## 5.3 Alarm Detection

Modifier-based logic enables severity classification and structured alarm generation.

## 5.4 Trip Detection

IEC-based deterministic logic ensures safe and controlled identification of shutdown conditions.

## 5.5 RAG Knowledge Integration

The structured Excel sheets can be converted into:

- JSON datasets
- Relational database entries
- Vector embeddings (e.g., FAISS)

This ensures that Generative AI reasoning remains constrained by engineering rules rather than probabilistic assumptions.

# 6. Importance of the Task

Without structured engineering rules, AI interpretation becomes probabilistic and potentially unsafe.

With structured deterministic logic:

- Safety is preserved
- Outputs become auditable
- Engineering alignment is ensured
- Scalability becomes feasible

This transformation shifts AI from pattern guessing to constrained engineering reasoning.

# 7. Scalability and Future Expansion

The current rule base:

- Covers representative instrument and safety combinations
- Is expandable using ISA combinational logic
- Can scale beyond 200 entries without structural redesign

Future enhancements may include:

- Expanded ISA variable coverage
- SIL-level classification
- Cause-and-effect matrix integration
- Interlock logic modeling
- Automated compliance validation

# 8. Conclusion

This task successfully:

- Translated industrial standards into structured engineering logic
- Built a deterministic knowledge base
- Prepared AI-ready rule sheets
- Ensured alignment with recognized safety standards

This work establishes the foundational engineering framework required for safe, explainable, and scalable AI-based extraction of E&I deliverables from P&ID drawings.